

Mathematical Statistics II Final Project

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Given Research Question: Is there a difference in study habits between traditional college students ages 18–24 and non-traditional college students ages 25 and older?

Variables Selected

Variable	Description
work_study_balance	Represents how students balance academics with work or outside responsibilities.
study_efficiency	Represents how productive or effective students perceive their study time to be.
library_visits_week	Represents frequency of library use and may reflect study environment, campus engagement, or reliance on library resources.

Together, these variables capture multiple aspects of study habits: time management and outside responsibilities, quality of studying, and study environment, or diligence. Each variable provides unique insight into a student's study habits. This is important because traditional and non-traditional students may differ in how they study, not just in how often they study. Also, these variables all provide diverse comparisons between traditional and non-traditional students.

Statistical Methods and Analysis Plan

The analysis was conducted at the 10% significance level. This project compares measures of study habits across students from several perspectives: center, spread, mean difference, direction of the difference, and distribution.

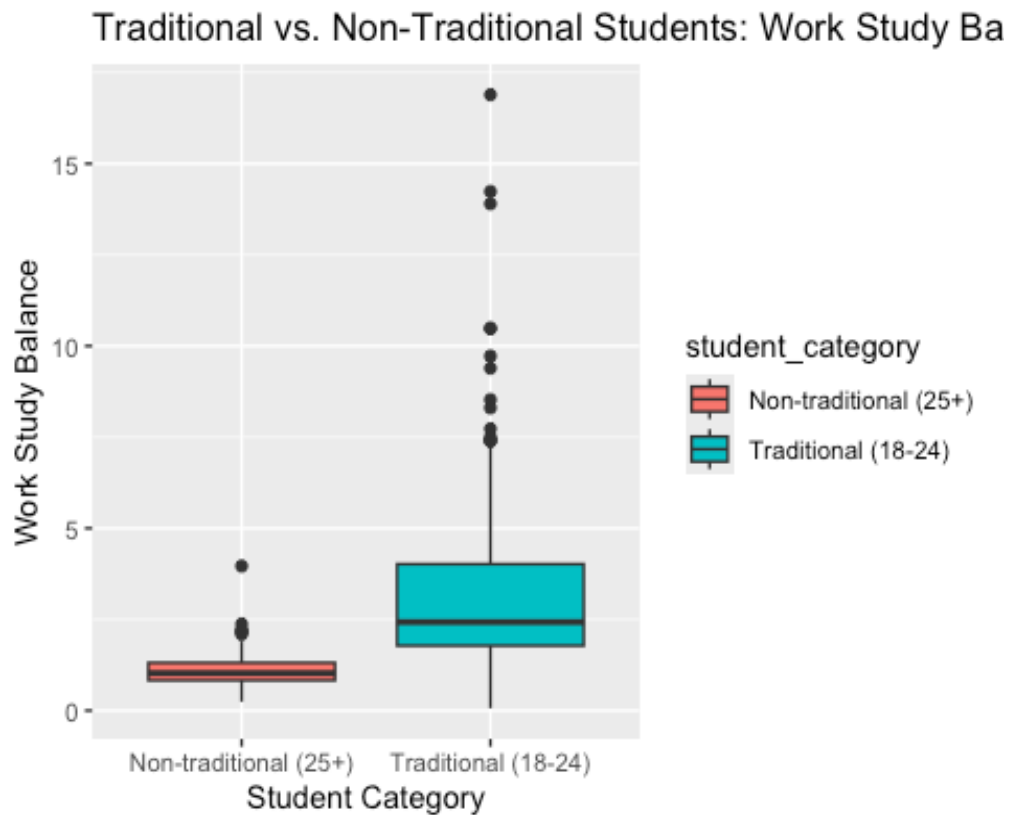
- Exploratory analysis used sample size, mean, median, variance, boxplots, and density plots.
- Variance tests were used to decide whether a pooled t-test or a Welch t-test was more appropriate for comparing means.
- Two-sided t-tests answered the main research question: whether the mean study habit measure differed between the two student categories.
- One-sided t-tests, 90% confidence intervals, and density plots were used to interpret the direction of the difference.
- Chi-square tests for equality of distributions compared whether traditional and non-traditional students were distributed similarly across low, middle, and high categories.

Exploratory Data Analysis

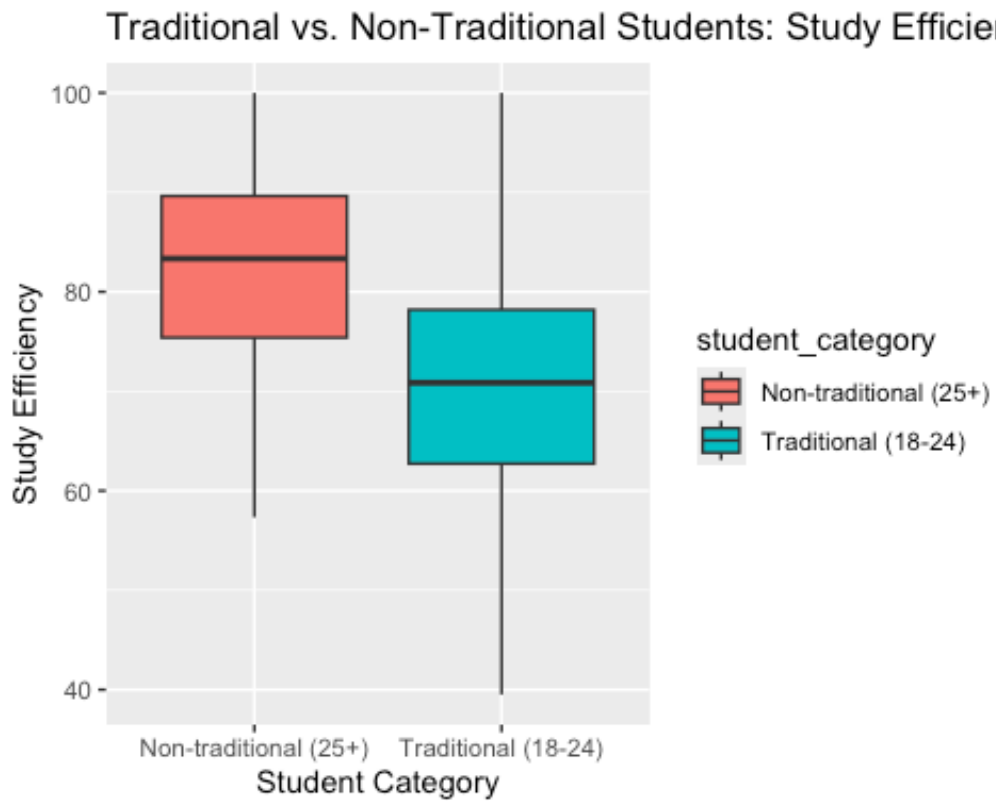
Before conducting hypothesis tests, I will separate the dataset into traditional and non-traditional students and perform exploratory data analysis for each variable. This step will allow me to understand the shape, center, and spread of the data, identify possible outliers, and support the choice of statistical procedure.

- For each variable and each group, the standard statistics will be calculated: sample size, sample mean, sample median, and sample standard deviation.
- To analyze the spread, skewness, and outliers of the variables between the two categories, comparative side-by-side boxplots will be used.

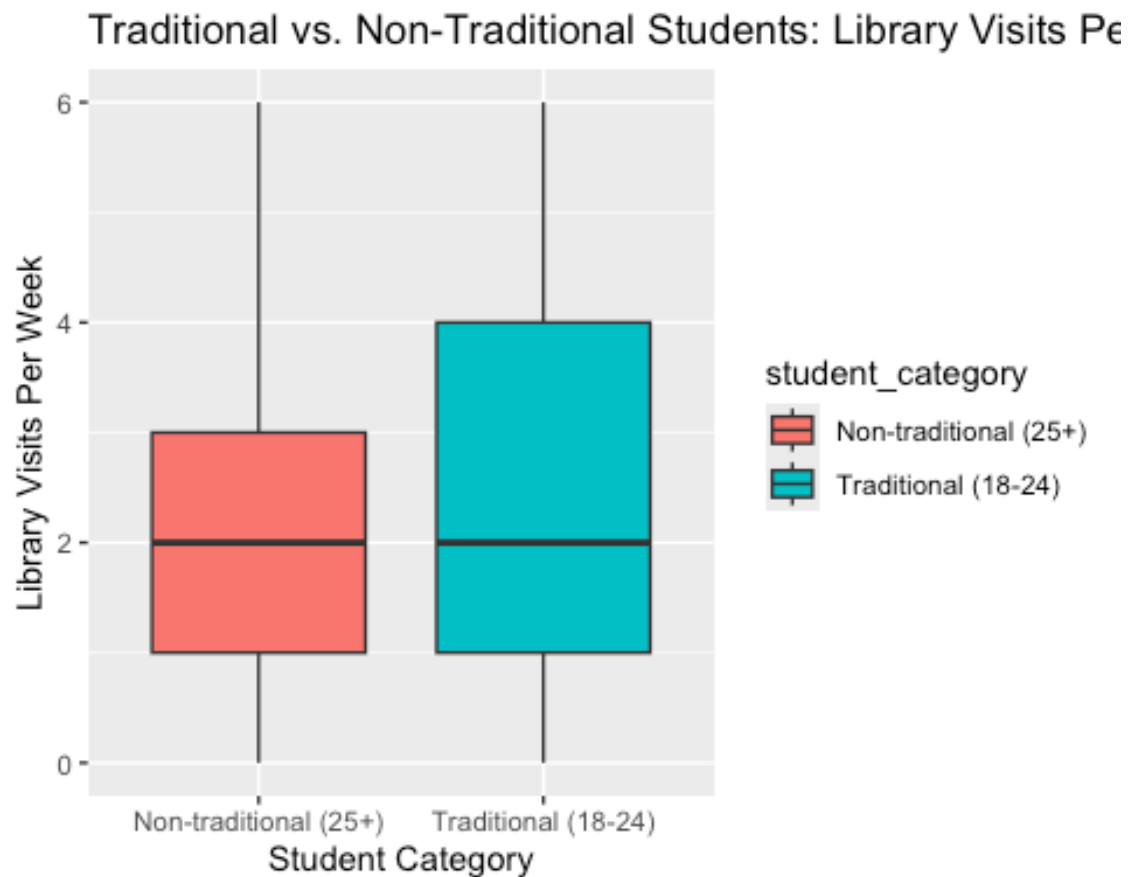
Group / Variable	N	Mean	Median	Variance
Traditional: work_study_balance	150	3.342	2.426	7.054
Non-traditional: work_study_balance	100	1.114	1.027	0.263
Traditional: study_efficiency	150	70.672	70.852	128.867
Non-traditional: study_efficiency	100	83.220	83.336	94.955
Traditional: library_visits_week	150	2.560	2.000	2.409
Non-traditional: library_visits_week	100	1.980	2.000	1.939



Work-study balance has the largest difference in both center and spread, with traditional students having a much larger mean and variance. There are also many outliers for traditional students, bringing the overall mean up. This is most likely due to students who are not working while in school.



Study efficiency also shows a clear difference in the center, with non-traditional students scoring higher on average. The distribution of the data looks the same, just around a different median.



Library visits per week are more subtle: both groups have the same median of 2 visits per week, but the traditional group has a higher mean. This suggests that the difference is not visible from the median alone, and the amount of library visits per week across student categories may not be as similar as they appear in the box plot.

The library visits variable was originally chosen as it seemed that the means would be similar. From the exploratory data analysis, I initially expected little or no difference in the means. However, the boxplot also shows that the traditional group has a wider third quartile.

Two-Tailed Hypothesis Test of Two Variances

Work Study Balance – Test of Two Variances

$$H_0: \sigma_t^2 = \sigma_n^2$$

$$H_1: \sigma_t^2 \neq \sigma_n^2$$

```
f_decision <- function(F_test_stat, F_crit) {
  cat("Test statistic is:", round(F_test_stat, 4), "\n")
  cat("Critical value is:", round(F_crit, 4), "\n")
}
```

```

if (F_test_stat > F_crit) {
  cat(
    "Reject the null hypothesis because the test statistic F =",
    round(F_test_stat, 4),
    "is greater than the critical value",
    round(F_crit, 4), ".\n"
  )
  cat("There is significant evidence to show that the population variances
are not equal, thus we will use Welch's t-test when testing the two means.\n"
)
} else {
  cat(
    "Fail to reject the null hypothesis because the test statistic F =",
    round(F_test_stat, 4),
    "is not greater than the critical value",
    round(F_crit, 4), ".\n"
  )
  cat("There is not significant evidence to show that the population varian
ces are not equal, thus we can use a pooled t-test when testing the two means
.\n")
}
}

S2_t_ws <- var(t_ws_balance)
S2_nt_ws <- var(nt_ws_balance)

n_t <- length(t_ws_balance)
n_nt <- length(nt_ws_balance)

F_test_stat <- (S2_t_ws/S2_nt_ws)
F_crit <- qf(0.95, n_t - 1, n_nt - 1)

f_decision(F_test_stat, F_crit)

## Test statistic is: 26.8045
## Critical value is: 1.361
## Reject the null hypothesis because the test statistic F = 26.8045 is great
er than the critical value 1.361.
## There is significant evidence to show that the population variances are no
t equal, thus we will use Welch's t-test when testing the two means.

```

The work-study balance variable required Welch's t-test because the traditional group had much more variability. This result agrees with the boxplot and density plot: traditional students show a wider range of work-study balance values, including several high observations. Study efficiency and library visits had variances close enough to support the pooled t-test.

Study Efficiency – Test of Two Variances

$$H_0: \sigma_t^2 = \sigma_n^2$$

$$H_1: \sigma_t^2 \neq \sigma_n^2$$

```
S2_t_se <- var(t_study_effic)
S2_nt_se <- var(nt_study_effic)

F_test_stat <- (S2_t_se/S2_nt_se)
F_crit <- qf(0.95, n_t - 1, n_nt - 1)

f_decision(F_test_stat, F_crit)

## Test statistic is: 1.3571
## Critical value is: 1.361
## Fail to reject the null hypothesis because the test statistic F = 1.3571 is
## not greater than the critical value 1.361.
## There is not significant evidence to show that the population variances are
## not equal, thus we can use a pooled t-test when testing the two means.
```

Library Visits Per Week – Test of Two Variances

$$H_0: \sigma_t^2 = \sigma_n^2$$

$$H_1: \sigma_t^2 \neq \sigma_n^2$$

```
S2_t_lv <- var(t_lib_visits)
S2_nt_lv <- var(nt_lib_visits)

F_test_stat <- (S2_t_lv/S2_nt_lv)
F_crit <- qf(0.95, n_t - 1, n_nt - 1)

f_decision(F_test_stat, F_crit)

## Test statistic is: 1.2425
## Critical value is: 1.361
## Fail to reject the null hypothesis because the test statistic F = 1.2425 is
## not greater than the critical value 1.361 .
## There is not significant evidence to show that the population variances are
## not equal, thus we can use a pooled t-test when testing the two means.
```

Variable	F statistic	Critical value	Decision	Test used
Work Study Balance	26.8045	1.3610	Reject equal variances	Welch's t-test
Study Efficiency	1.3571	1.3610	Fail to reject equal variances	Pooled t-test
Library Visits Per Week	1.2425	1.3610	Fail to reject equal variances	Pooled t-test

Two-Tailed Hypothesis Test for Two Means

Work Study Balance - Welch's T-Test

$$H_0: \mu_t = \mu_n$$

$$H_1: \mu_t \neq \mu_n$$

```
t_decision <- function(variable, test_stat, df, alpha, alternative, test_type) {  
  
  if (alternative == "not equal to") {  
    t_crit_right <- qt(1 - alpha / 2, df)  
    t_crit_left <- -t_crit_right  
    reject <- abs(test_stat) > t_crit_right  
  
    p_value <- 2 * pt(-abs(test_stat), df)  
  
    cat("\n", test_type, "-", variable, "\n")  
    cat("Test statistic:", round(test_stat, 4), "\n")  
    cat("Degrees of freedom:", round(df, 4), "\n")  
    cat("Critical values: ±", round(t_crit_right, 4), "\n")  
    cat("Each rejection region has area alpha/2 =", alpha / 2, "\n")  
    # Will show up as 0 when using regular round(p_value, 6) if it is very sm  
all.  
    cat("P-value:", format.pval(p_value, digits = 6, eps = .Machine$double.xm  
in), "\n")  
  }  
  
  else if (alternative == "greater than") {  
    t_crit_right <- qt(1 - alpha, df)  
    t_crit_left <- NA  
    reject <- test_stat > t_crit_right  
  
    p_value <- pt(test_stat, df, lower.tail = FALSE)  
  
    cat("\n", test_type, "-", variable, "\n")  
    cat("Test statistic:", round(test_stat, 4), "\n")  
    cat("Degrees of freedom:", round(df, 4), "\n")  
    cat("Right-tail critical value:", round(t_crit_right, 4), "\n")  
    cat("Rejection region area = alpha =", alpha, "\n")  
    # Will show up as 0 when using regular round(p_value, 6) if it is very sm  
all.  
    cat("P-value:", format.pval(p_value, digits = 6, eps = .Machine$double.xm  
in), "\n")  
  }  
}
```

```

else if (alternative == "less than") {
  t_crit_left <- qt(alpha, df)
  t_crit_right <- NA
  reject <- test_stat < t_crit_left

  p_value <- pt(test_stat, df, lower.tail = TRUE)

  cat("\n", test_type, "-", variable, "\n")
  cat("Test statistic:", round(test_stat, 4), "\n")
  cat("Degrees of freedom:", round(df, 4), "\n")
  cat("Left-tail critical value:", round(t_crit_left, 4), "\n")
  cat("Rejection region area = alpha =", alpha, "\n")
  # Will show up as 0 when using regular round(p_value, 6) if it is very sm
all.
  cat("P-value:", format.pval(p_value, digits = 6, eps = .Machine$double.xm
in), "\n")
}

else {
  stop("alternative must be 'not equal to', 'greater than', or 'less than'"
)
}

if (reject) {
  cat("Reject the null hypothesis.\n")
  cat("There is significant evidence that the mean of traditional students'
",
      variable, "is", alternative, "the mean of non-traditional students.\n
")
} else {
  cat("Fail to reject the null hypothesis.\n")
  cat("There is not enough evidence to support that the mean of traditional
students'",
      variable, "is", alternative, "the mean of non-traditional students.\n
")
}

return(data.frame(
  variable = variable,
  test_type = test_type,
  test_stat = test_stat,
  df = df,
  alpha = alpha,
  alternative = alternative,
  t_crit_left = t_crit_left,
  t_crit_right = t_crit_right,
  p_value = p_value,
  reject = reject

```

```

))
}

welch_t_test <- function(traditional_var, nontraditional_var, null_diff, alpha=0.10){

  x_bar <- mean(traditional_var)
  y_bar <- mean(nontraditional_var)

  x_s2 <- var(traditional_var)
  y_s2 <- var(nontraditional_var)

  nx <- length(traditional_var)
  ny <- length(nontraditional_var)

  test_stat <- ((x_bar - y_bar) - null_diff) / sqrt((x_s2 / nx) + (y_s2 / ny)
)

  r <- (((x_s2 / nx) + (y_s2 / ny)) ^ 2) / (((1/(nx - 1)) * (x_s2 / nx ) ^ 2
) + ((1/(ny - 1)) * (y_s2 / ny ) ^ 2 ))
  r <- floor(r)

  t_crit <- qt(1 - alpha / 2, r)

  ci_lower <- (x_bar - y_bar) - t_crit * sqrt((x_s2 / nx) + (y_s2 / ny))
  ci_upper <- (x_bar - y_bar) + t_crit * sqrt((x_s2 / nx) + (y_s2 / ny))

  return(list(t = test_stat, df = r, lower = ci_lower, upper = ci_upper))
}

work_study_result <- welch_t_test(t_ws_balance, nt_ws_balance, 0)
work_study_t <- work_study_result$t
work_study_df <-work_study_result$df

ws_two_sided <- t_decision("work study balance", work_study_t, work_study_df,
0.10, "not equal to", "Two-Sided")

##
## Two-Sided - work study balance
## Test statistic: 10.0002
## Degrees of freedom: 165
## Critical values: ± 1.6541
## Each rejection region has area alpha/2 = 0.05
## P-value: 1.05695e-18
## Reject the null hypothesis.
## There is significant evidence that the mean of traditional students' work
study balance is not equal to the mean of non-traditional students.

```

Study Efficiency - Pooled T-Test

$$H_0: \mu_t = \mu_n$$

$$H_1: \mu_t \neq \mu_n$$

```
pooled_t_test <- function(traditional_var, nontraditional_var, null_diff, alpha = 0.10){
  x_bar <- mean(traditional_var)
  y_bar <- mean(nontraditional_var)

  nx <- length(traditional_var)
  ny <- length(nontraditional_var)

  sx2 <- var(traditional_var)
  sy2 <- var(nontraditional_var)

  sp2 <- (((nx-1) * sx2) + ((ny-1) * sy2)) / (nx + ny - 2)

  t_test_stat <- ((x_bar - y_bar) - null_diff) / sqrt(sp2 * ((1/nx)+(1/ny)))
  df <- (nx + ny - 2)

  t_crit <- qt(1 - alpha / 2, df)

  ci_lower <- (x_bar - y_bar) - t_crit * sqrt(sp2 * ((1/nx)+(1/ny)))
  ci_upper <- (x_bar - y_bar) + t_crit * sqrt(sp2 * ((1/nx)+(1/ny)))

  return(list(t = t_test_stat, df = df, lower = ci_lower, upper = ci_upper))
}

study_effic_pooled_result <- pooled_t_test(t_study_effic, nt_study_effic, 0)
study_effic_t_stat <- study_effic_pooled_result$t
study_effic_df <- study_effic_pooled_result$df

se_two_sided <- t_decision("study efficiency", study_effic_t_stat, study_effic_df, 0.10, "not equal to", "Two-Sided")

##
## Two-Sided - study efficiency
## Test statistic: -9.0506
## Degrees of freedom: 248
## Critical values: ± 1.651
## Each rejection region has area alpha/2 = 0.05
## P-value: 4.28621e-17
## Reject the null hypothesis.
## There is significant evidence that the mean of traditional students' study efficiency is not equal to the mean of non-traditional students.
```

Library Visits Per Week - Pooled T-Test

$$H_0: \mu_t = \mu_n$$

$$H_1: \mu_t \neq \mu_n$$

```
library_pooled_result <- pooled_t_test(t_lib_visits, nt_lib_visits, 0)
lib_t_stat <- library_pooled_result$t
lib_df <- library_pooled_result$df

lv_two_sided <- t_decision("library visits per week", lib_t_stat, lib_df, 0.1
0, "not equal to", "Two-Sided")

##
## Two-Sided - library visits per week
## Test statistic: 3.0143
## Degrees of freedom: 248
## Critical values: ± 1.651
## Each rejection region has area alpha/2 = 0.05
## P-value: 0.00284306
## Reject the null hypothesis.
## There is significant evidence that the mean of traditional students' libra
ry visits per week is not equal to the mean of non-traditional students.
```

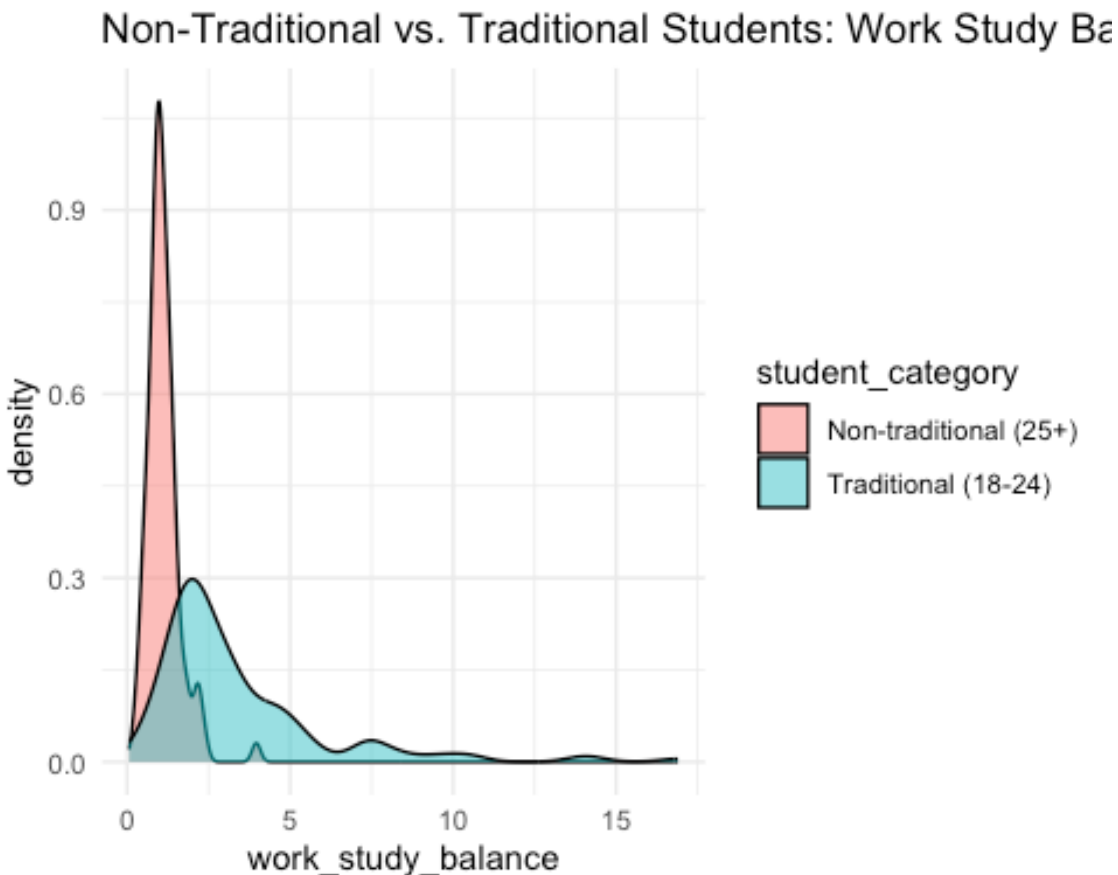
Variable	Test	t statistic	Df	p-value	Decision
Work Study Balance	Welch	10.0002	165	1.05695e-18	Reject H0
Study Efficiency	Pooled	-9.0506	248	4.28621e-17	Reject H0
Library Visits Per Week	Pooled	3.0143	248	0.00284306	Reject H0

The two-sided tests answer the main research question. For each variable, the p-value is less than 0.10, so there is statistically significant evidence that the true means of the variables representing different study habit aspects are all different across traditional and non-traditional students. Confidence intervals and one-sided tests are needed to interpret the direction.

Exactly How do the Means Differ Between Student Category?

Viewing Side-By-Side Distributions

Traditional Students Have Higher Work Study Balance



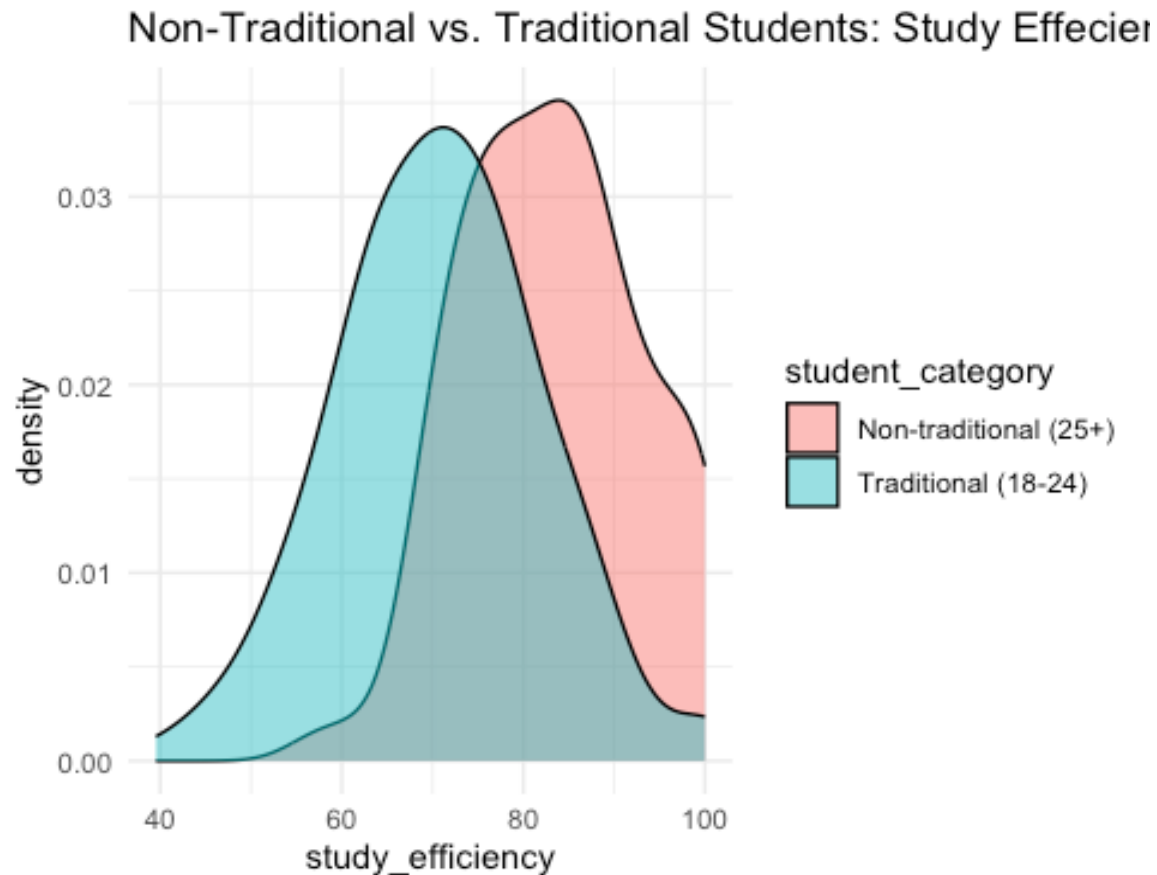
```
## 90% Confidence Interval for Traditional vs. Non-Traditions students' work  
study balance: [ 1.859825 , 2.597038 ].
```

```
## The confidence interval does not contain zero, which agrees with the state  
ment that the means of work study balance are different across student catego  
ry.
```

```
## The confidence interval contains only positive numbers, indicating that tr  
aditional students have a higher work study balance than non-traditional stud  
ents.
```

You can see how the outliers in traditional students affect this density plot. It is much wider than the plot for non-traditional students.

Non-Traditional Students Have a Higher Study Efficiency

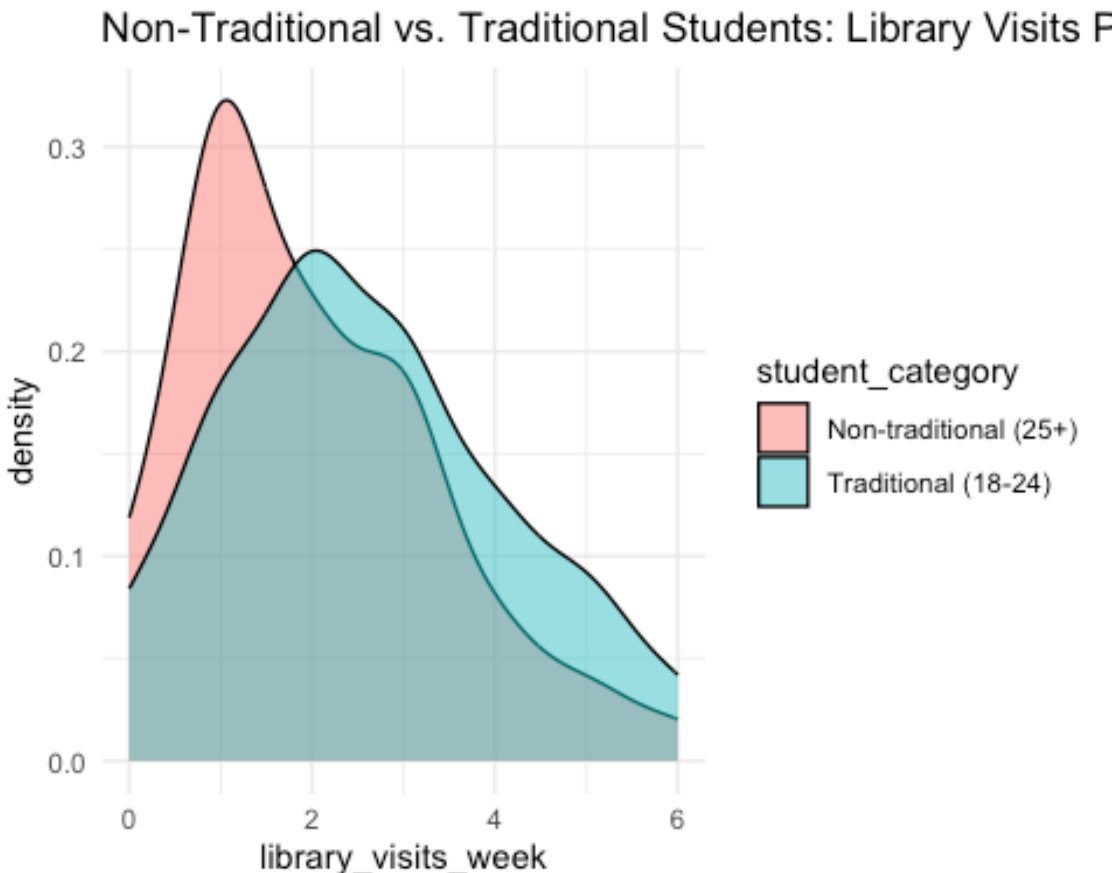


```
## 90% Confidence Interval for Traditional vs. Non-Traditions students' study  
efficiency: [ -14.83698 , -10.25896 ].
```

```
## The confidence interval does not contain zero, which agrees with the state  
ment that the means of study efficiency are different across student category  
.
```

```
## The confidence interval contains only negative numbers, indicating that tr  
aditional students have a lower study efficiency than non-traditional student  
s.
```

Traditional Students Visit The Library More Times Per Week



90% Confidence Interval for Traditional vs. Non-Traditions students' library visits per week: [0.262316 , 0.897684].

The confidence interval does not contain zero, which agrees with the statement that the means of library visits per week are different across student category.

The confidence interval contains only positive numbers, indicating that traditional students visit the library more frequently in a week than non-traditional students.

Variable	90% CI for $\mu_t - \mu_n$	Interpretation
Work Study Balance	[1.860, 2.597]	Traditional students have a higher mean work-study balance score.
Study Efficiency	[-14.837, -10.259]	Non-traditional students have a higher mean study efficiency score.
Library Visits Per Week	[0.262, 0.898]	Traditional students visit the library more times per week on average.

Official One-Sided Hypothesis Tests of Two Means

Work Study Balance - Welch Test

$$H_0: \mu_t = \mu_n$$

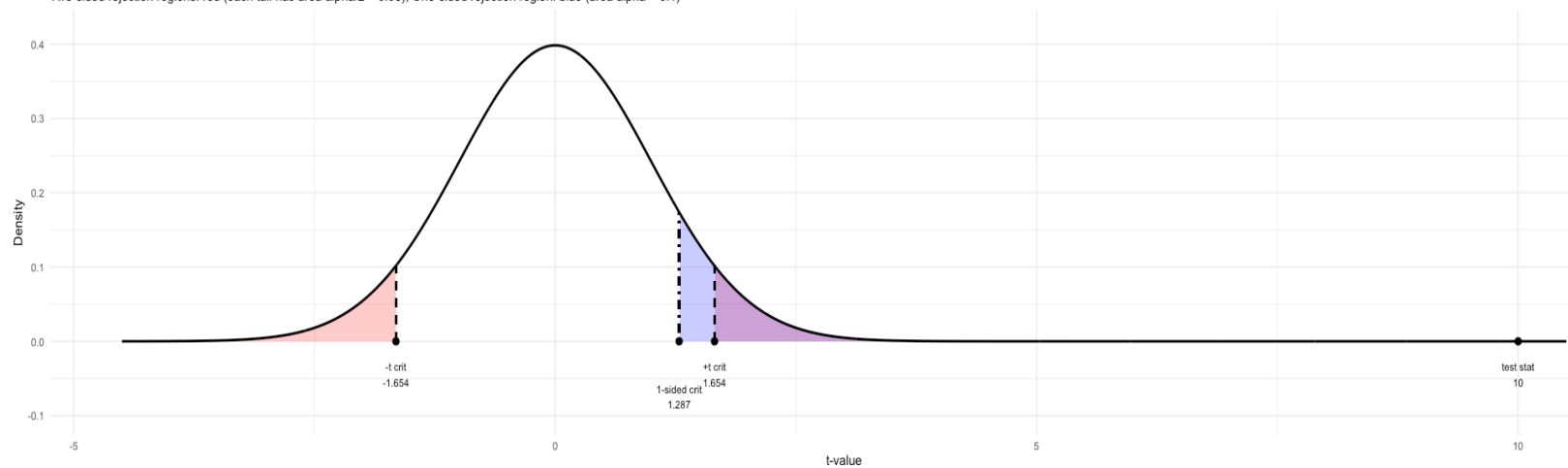
$$H_1: \mu_t > \mu_n$$

```
# same test-stat and degrees of freedom as the two-tailed test
ws_one_sided <- t_decision("work study balance", work_study_t, work_study_df,
0.10, "greater than", "One-Sided")

##
## One-Sided - work study balance
## Test statistic: 10.0002
## Degrees of freedom: 165
## Right-tail critical value: 1.2867
## Rejection region area = alpha = 0.1
## P-value: 5.28476e-19
## Reject the null hypothesis.
## There is significant evidence that the mean of traditional students' work
study balance is greater than the mean of non-traditional students.
```

Work Study Balance: T-Distribution

Two-sided rejection regions: red (each tail has area $\alpha/2 = 0.05$); One-sided rejection region: blue (area $\alpha = 0.1$)



Study Efficiency - Pooled T-Test

$$H_0: \mu_t = \mu_n$$

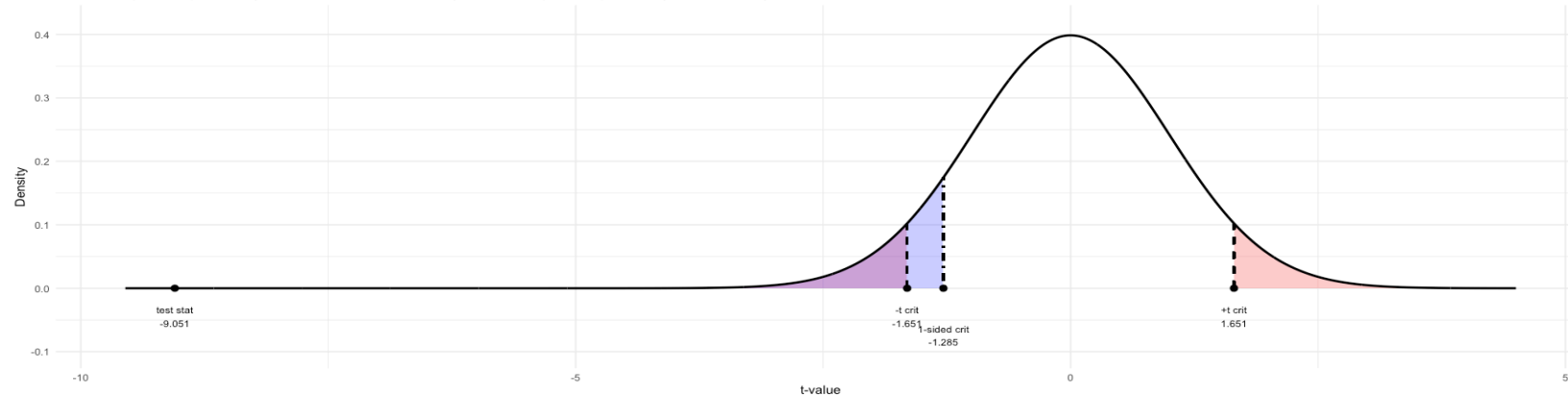
$$H_1: \mu_t < \mu_n$$

```
se_one_sided <- t_decision("study efficiency", study_effic_t_stat, study_effi
c_df, 0.10, "less than", "One-Sided")

##
## One-Sided - study efficiency
```

Study Efficiency: T-Distribution

Two-sided rejection regions: red (each tail has area alpha/2 = 0.05); One-sided rejection region: blue (area alpha = 0.1)



```
## Test statistic: -9.0506
## Degrees of freedom: 248
## Left-tail critical value: -1.285
## Rejection region area = alpha = 0.1
## P-value: 2.14311e-17
## Reject the null hypothesis.
## There is significant evidence that the mean of traditional students' study
efficiency is less than the mean of non-traditional students.
```

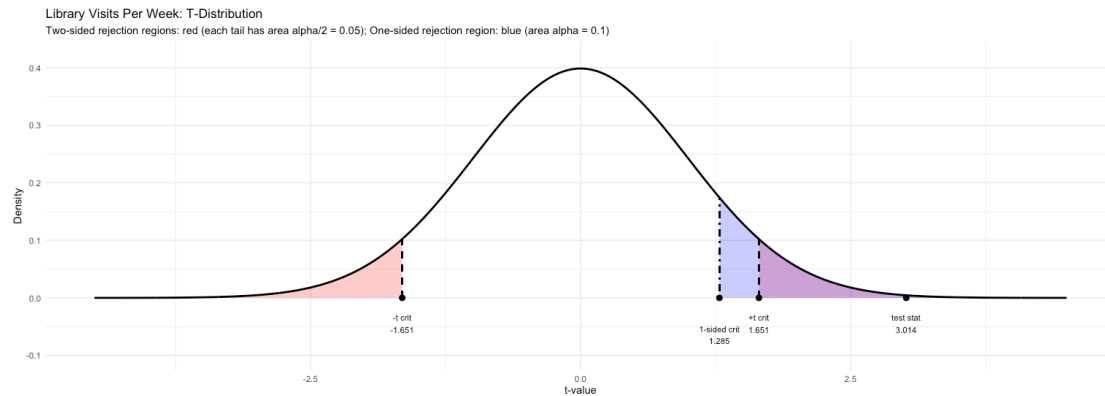
Library Visits Per Week - Pooled T-Test

$$H_0: \mu_t = \mu_n$$

$$H_1: \mu_t > \mu_n$$

```
lv_one_sided <- t_decision("library visits per week", lib_t_stat, lib_df, 0.1
0, "greater than", "One-Sided")

##
## One-Sided - library visits per week
## Test statistic: 3.0143
## Degrees of freedom: 248
## Right-tail critical value: 1.285
## Rejection region area = alpha = 0.1
## P-value: 0.00142153
## Reject the null hypothesis.
## There is significant evidence that the mean of traditional students' libra
ry visits per week is greater than the mean of non-traditional students.
```



Hypothesis Testing: Homogeneity

Looking Back at The Density Plots - Comparing the Distributions of Each Variable Across Study Categories Could Provide Meaningful Insight

For each variable, the traditional and non-traditional student data were combined. That data was ordered and divided into three separate groups. The ends of each group create bounds for low, middle, and high.

After the variable is separated again by category, we will count how many values fall into low, middle, and high.

```
distribution_homogeneity_test <- function(variable_name, x, group, alpha = 0.10, num_groups = 3) {  
  
  # Remove missing values  
  clean_data <- data.frame(  
    x = x,  
    group = group  
  )  
  
  clean_data <- clean_data[!is.na(clean_data$x) & !is.na(clean_data$group), ]  
  
  # Create distribution categories using combined-sample quantiles  
  probs <- seq(0, 1, length.out = num_groups + 1)  
  breaks <- quantile(clean_data$x, probs = probs, na.rm = TRUE)  
  
  # Remove duplicate break points if needed  
  breaks <- unique(breaks)  
  
  # Make sure all values are included  
  breaks[1] <- -Inf  
  breaks[length(breaks)] <- Inf
```

```

# Create group labels
if ((length(breaks) - 1) == 3) {
  labels <- c("Low", "Middle", "High")
} else if ((length(breaks) - 1) == 4) {
  labels <- c("Low", "Lower-Middle", "Upper-Middle", "High")
} else {
  labels <- paste0("Group ", 1:(length(breaks) - 1))
}

clean_data$distribution_group <- cut(
  clean_data$x,
  breaks = breaks,
  include.lowest = TRUE,
  labels = labels
)

# Observed contingency table
observed <- table(clean_data$group, clean_data$distribution_group)

# Row totals, column totals, and grand total
row_totals <- rowSums(observed)
column_totals <- colSums(observed)
grand_total <- sum(observed)

# Expected counts
expected <- outer(row_totals, column_totals) / grand_total

# Chi-square statistic
Q_stat <- sum((observed - expected)^2 / expected)

# Degrees of freedom
r <- nrow(observed)
c <- ncol(observed)
df <- (r - 1) * (c - 1)

# Critical value and p-value
Q_crit <- qchisq(1 - alpha, df)
p_value <- pchisq(Q_stat, df, lower.tail = FALSE)

reject <- Q_stat > Q_crit

cat("Observed counts:\n")
print(observed)

cat("\nExpected counts:\n")
print(round(expected, 3))

cat("\nQ statistic:", round(Q_stat, 4), "\n")

```

```

cat("Critical value:", round(Q_crit, 4), "\n")
cat("Degrees of freedom:", df, "\n")
cat("P-value:", format.pval(p_value, digits = 6, eps = .Machine$double.xmin
), "\n")

if (reject) {
  cat("\nReject the null hypothesis.\n")
  cat("There is significant evidence that the distribution of", variable_name,
      "differs across student categories.\n")
} else {
  cat("\nFail to reject the null hypothesis.\n")
  cat("There is not significant evidence that the distribution of", variable_name,
      "differs across student categories.\n")
}

return(list(
  variable = variable_name,
  observed = observed,
  expected = expected,
  Q_stat = Q_stat,
  Q_crit = Q_crit,
  df = df,
  alpha = alpha,
  p_value = p_value,
  reject = reject,
  clean_data = clean_data
))
}

```

Work Study Balance - Chi-Square Homogeneity Test

$$H_0: p_{ti} = p_{ni}, \quad i = 1, 2, 3$$

$$H_1: p_{ti} \neq p_{ni} \text{ for at least one } i$$

```

ws_dist_test <- distribution_homogeneity_test(
  variable_name = "Work Study Balance",
  x = study_data$work_study_balance,
  group = study_data$student_category,
  alpha = 0.10
)

```

Observed counts:

```

##
##               Low Middle High
## Non-traditional (25+) 69    29    2
## Traditional (18-24)  15    54   81
##

```

```
## Expected counts:
##               Low Middle High
## Non-traditional (25+) 33.6   33.2 33.2
## Traditional (18-24)   50.4   49.8 49.8
##
## Q statistic: 111.9137
## Critical value: 4.6052
## Degrees of freedom: 2
## P-value: 4.99164e-25
##
## Reject the null hypothesis.
## There is significant evidence that the distribution of Work Study Balance
differs across student categories.
```

Study Efficiency - Chi-Square Homogeneity Test

$$H_0: p_{ti} = p_{ni}, \quad i = 1, 2, 3$$

$$H_1: p_{ti} \neq p_{ni} \text{ for at least one } i$$

```
se_dist_test <- distribution_homogeneity_test(
  variable_name = "Study Efficiency",
  x = study_data$study_efficiency,
  group = study_data$student_category,
  alpha = 0.10
)

## Observed counts:
##               Low Middle High
## Non-traditional (25+) 10     32  58
## Traditional (18-24)   74     51  25
##
## Expected counts:
##               Low Middle High
## Non-traditional (25+) 33.6   33.2 33.2
## Traditional (18-24)   50.4   49.8 49.8
##
## Q statistic: 58.5748
## Critical value: 4.6052
## Degrees of freedom: 2
## P-value: 1.90831e-13
##
## Reject the null hypothesis.
## There is significant evidence that the distribution of Study Efficiency di
ffers across student categories.
```

Library Visits Per Week - Homogeneity Test

$$H_0: p_{ti} = p_{ni}, \quad i = 1, 2, 3$$

$$H_1: p_{ti} \neq p_{ni} \text{ for at least one } i$$

```
lv_dist_test <- distribution_homogeneity_test(
  variable_name = "Library Visits Per Week",
  x = study_data$library_visits_week,
  group = study_data$student_category,
  alpha = 0.10,
  num_groups = 3
)

## Observed counts:
##
##               Low Middle High
## Non-traditional (25+) 46     41  13
## Traditional (18-24)  40     71  39
##
## Expected counts:
##               Low Middle High
## Non-traditional (25+) 34.4  44.8 20.8
## Traditional (18-24)  51.6  67.2 31.2
##
## Q statistic: 11.9316
## Critical value: 4.6052
## Degrees of freedom: 2
## P-value: 0.00256501
##
## Reject the null hypothesis.
## There is significant evidence that the distribution of Library Visits Per
Week differs across student categories.
```

Variable	Q statistic	Critical value	df	p-value	Decision
Work Study Balance	111.9137	4.6052	2	4.99164e-25	Reject H0
Study Efficiency	58.5748	4.6052	2	1.90831e-13	Reject H0
Library Visits Per Week	11.9316	4.6052	2	0.00256501	Reject H0
Variable	Traditional distribution		Non-traditional distribution		Interpretation
Work Study Balance	Low: 15, Middle: 54, High: 81		Low: 69, Middle: 29, High: 2		Traditional students are much more likely

			to be in the high category; non-traditional students are concentrated in the low category.
Study Efficiency	Low: 74, Middle: 51, High: 25	Low: 10, Middle: 32, High: 58	Non-traditional students are much more likely to be in the high study-efficiency category.
Library Visits Per Week	Low: 40, Middle: 71, High: 39	Low: 46, Middle: 41, High: 13	Even with equal medians, traditional students appear more often in the high library-visit category.

These hypothesis tests for homogeneity support the same overall conclusion as the t-tests and add an extra layer of information. The groups not only differ in means, but they are distributed differently across the range of possible values. These results are especially helpful for library visits, where the median alone made the groups look similar. The distribution test shows that the similarity at the median hides meaningful differences in the lower and upper parts of the distribution.

Overall Reflection and Conclusion

The original research question asked whether study habits differ between traditional and non-traditional students. Based on the selected variables, the answer is yes. However, the differences are not all in the same direction. Traditional students had higher work-study balance scores and more library visits per week, while non-traditional students had higher study efficiency scores.

It is important to analyze different aspects of the variables. The library visits variable shows why these matters. The medians were identical, so a quick comparison of medians might suggest no difference. The means, confidence interval, and chi-square distribution test revealed that the groups were different because the counts were spread differently across low, middle, and high visit levels. This made the result more meaningful than simply saying one group visits the library more often.

The study efficiency result also shows the value of using multiple methods. Visually, the density curves had a similar general shape, but the non-traditional group was shifted to higher values. The t-test captured the difference in means, while the chi-square test confirmed that the shift was large enough to change the distribution across low, middle, and high categories.

The results suggest that traditional and non-traditional students may study differently because their lives are structured differently. Traditional students may have more opportunities to be physically present on campus and use campus resources such as the library. Non-traditional

students may have less time available, so they may use that time more efficiently. Therefore, these results should not be interpreted as one group having better study habits overall. Instead, they show that study habits can differ in form: frequency, efficiency, and balance capture different dimensions of student behavior.